Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech (EE) Examination

May 2018

EE 745 FACTS and HVDC TRANSMISSION

Date: 03/05/2018, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the following in brief: **[08]**

a. What are the objectives of FACTS controllers?

b. List the disadvantage of fixed series compensation.

c. How is system voltage stability limit improved?

d. List the advantages of TCSC.

Q – 2.a State the types of compensation by function and type in tabular form. **[03]**

Q – 2.b Answer any two questions. **[10]**

(i) Explain working principle and V-I characteristics of Thyristor Controlled Reactor (TCR).

(ii) Explain the operating principle of UPFC with necessary diagram.

(iii) Derive the expression of degree of series and shunt compensation.

Q - 3 Answer any two questions. **[14]**

a. Explain the working principle and V-I characteristics of Static Synchronous Compensator (STATCOM) with suitable vector diagram.

b. Explain the working principle and V-I characteristics of Static VAR compensator (SVC). Also State the applications of SVC.

c. What are the characteristics of an uncompensated transmission line? Explain load compensation and system compensation w.r.t uncompensated transmission line.

SECTION – II

Q – 4.a **Choose the correct answer.** **[04]**

(i) With increasing in delay angle α

a) power factor is reduced

b) DC voltage decreases

c) both (a) and (b)

d) kVAR requirement decreases

(ii) During commutation in a converter

a) voltage is exchanged

b) current is transformed from one valve to the other

c) Dc voltage is blocked

d) none of the above

(iii) In a bipolar system

a) both conductor are positive

b) both conductor are negative

c) one conductor is positive and the other negative

d) one conductor is positive or negative and other is at ground potential

(iv) If pulse number = p, and k is an integer, voltage harmonic generated on the DC side is

a) pk+1

b) pk-1

c) 2pk

d) pk

Q – 4.b Explain the difference between active and passive compensator? **[03]**

Q – 5.a List the various types of HVDC links and explain them in brief. **[06]**

OR

Q – 5.a Explain with block diagram the current control system with equidistant pulse control? **[06]**

Q – 5.b Derive the expression of average direct no load voltage V_{d0} for a three phase six pulse rectifier with ignition delay α. **[04]**

Q – 5.c Explain the system control hierarchy for HVDC system controls. **[05]**

Q – 6.a Calculate the secondary line voltage of the transformer for 3-phase bridge rectifier to provide a dc voltage of 120 KV. Assume α=30˚, μ=15˚. What is the effective reactance, if the rectifier gives 800 Amp. of dc output current? **[04]**

Q – 6.b What are characteristic and non-characteristic harmonics? **[04]**

Q – 6.c A bipolar two terminal HVDC link is delivering 1000 MW at ±500 KV at the receiving end. The total losses in the dc circuit are 50 MW. Calculate the following: **[05]**

1. Sending end power
2. Sending end voltage
3. Power in the middle of the line
4. Voltage in the middle of the line
5. Total resistance of the DC circuit.

OR

Q – 6.c Summarize the basic control principles used for HVDC control. **[05]**

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